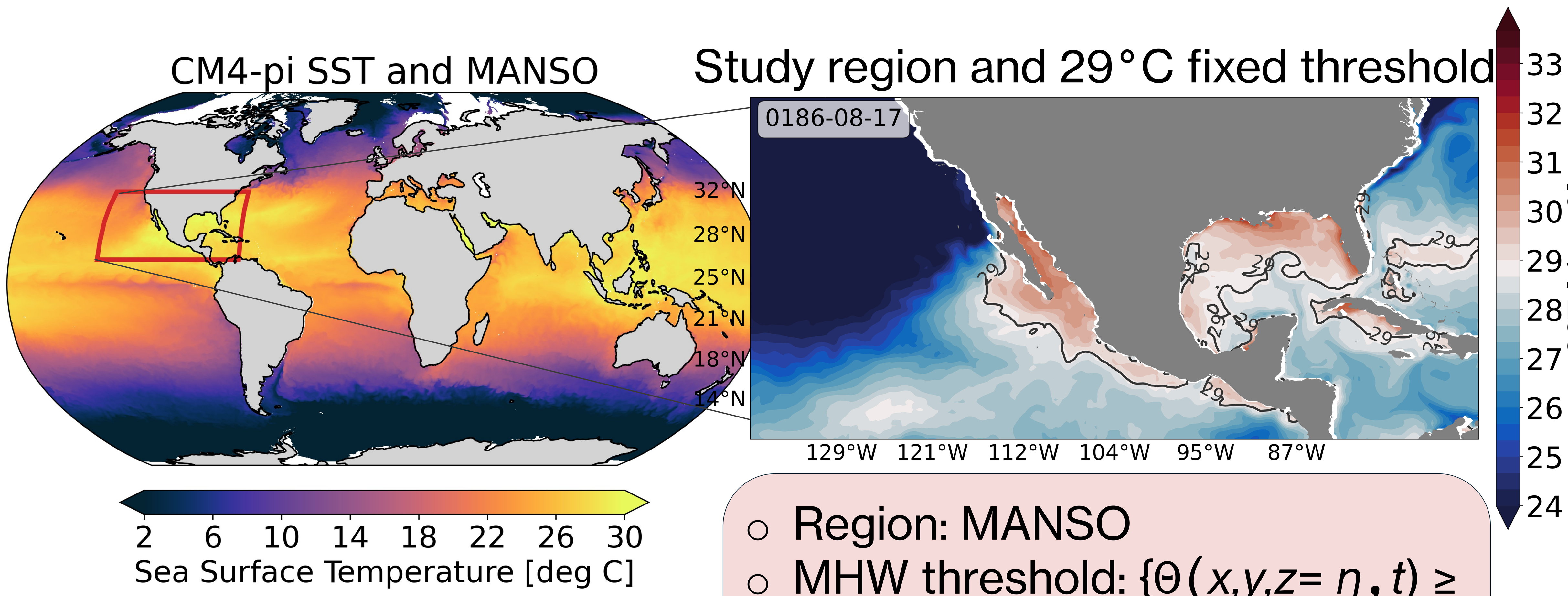
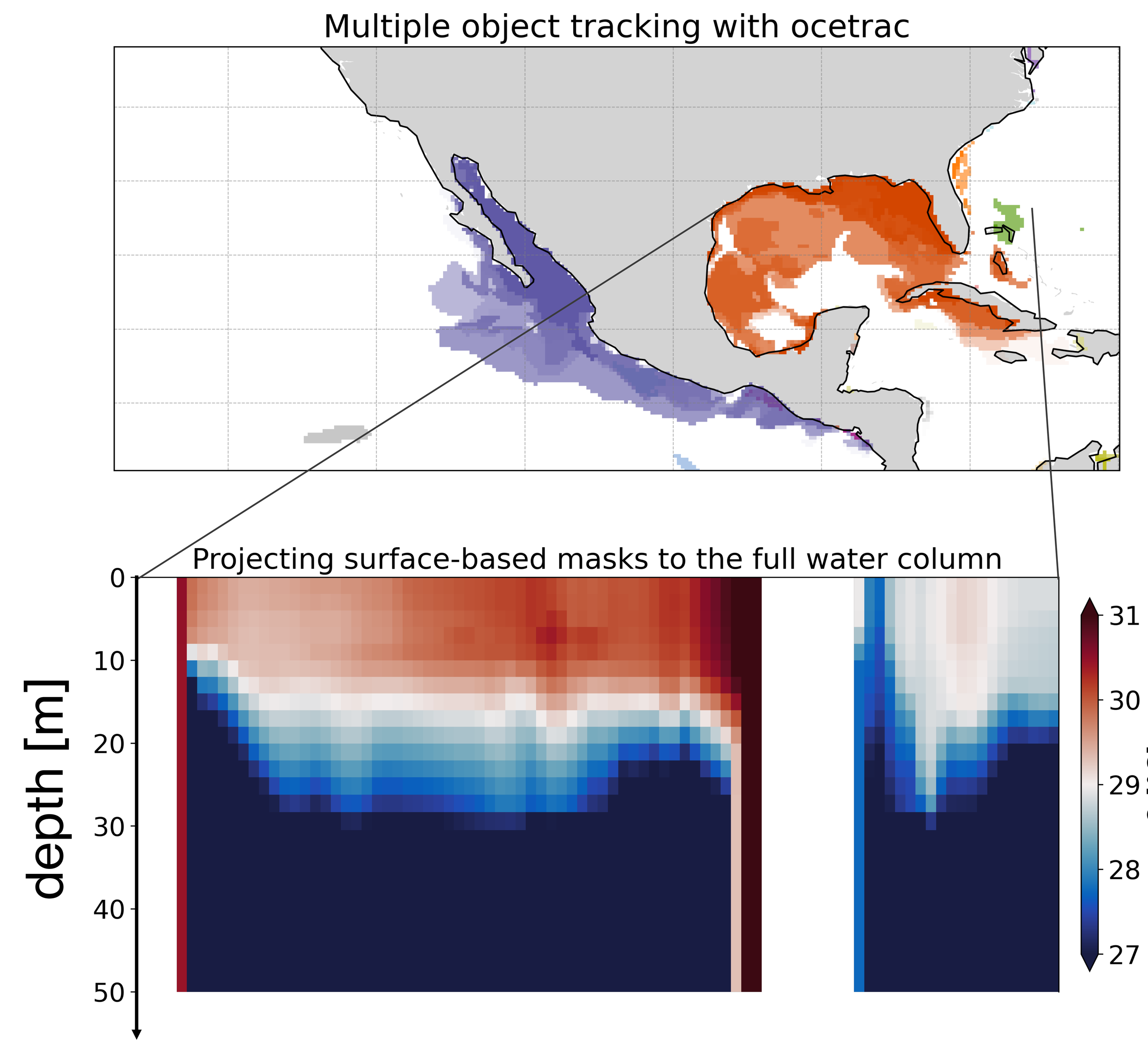
The Warm Water Mass Transformation Recipe: diagnosing dynamic mass budgets for individual object-tracked marine heatwaves

[1] Select a region (could be global), save diagnostics, select marine heatwave threshold



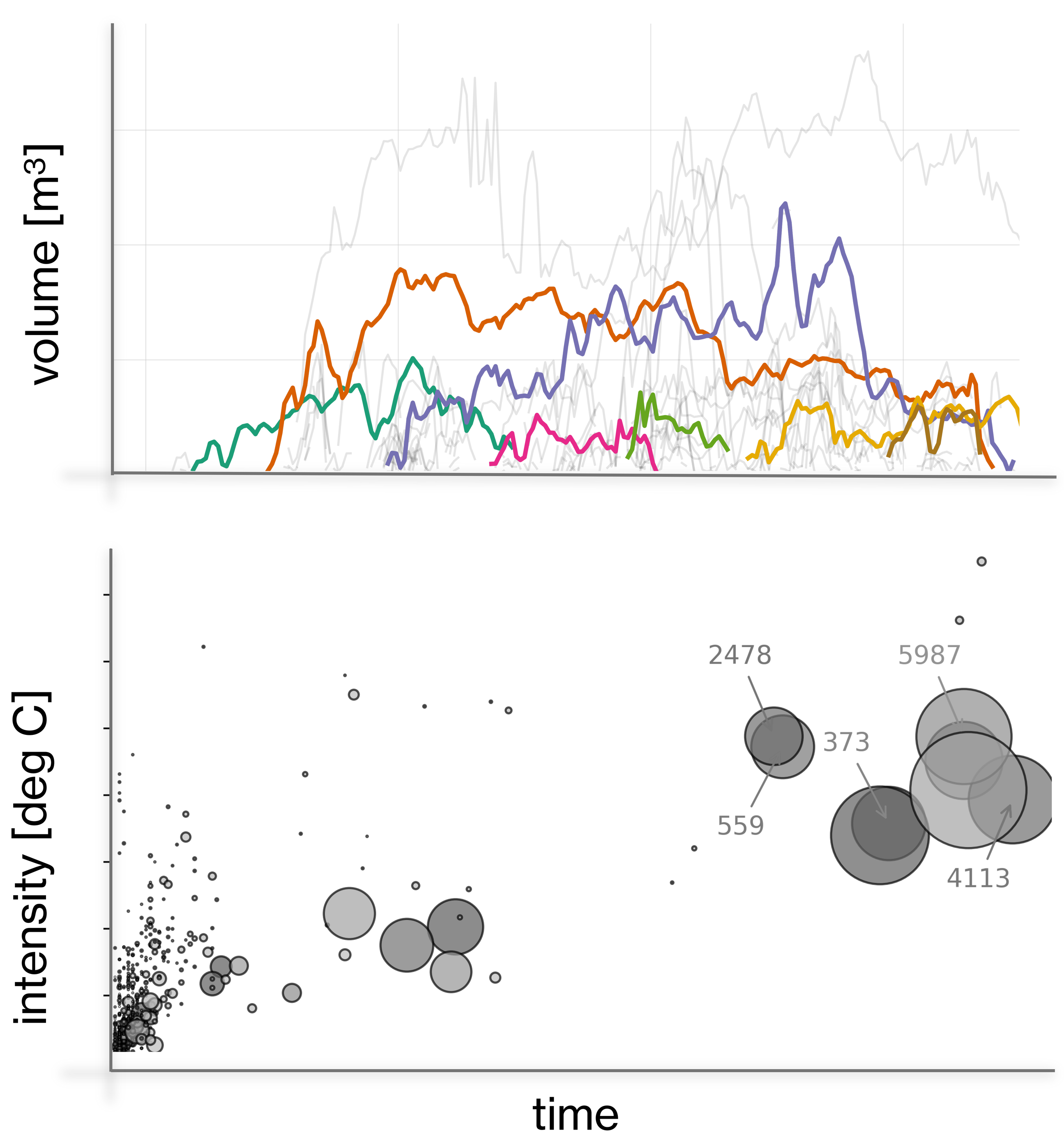
- Region: MANSO
- MHW threshold: $\{\Theta(x,y,z=\eta,t) \geq 29°C\}$ (gray contour)
- Save full-column daily tracer budget terms

[2a] Spatiotemporally coherent tracking and subsurface projections



- Set up extreme event tracker (ocetrac)
- From SST fields, track unique MHWs
- Project each tracked MHW mask vertically for subsurface structure

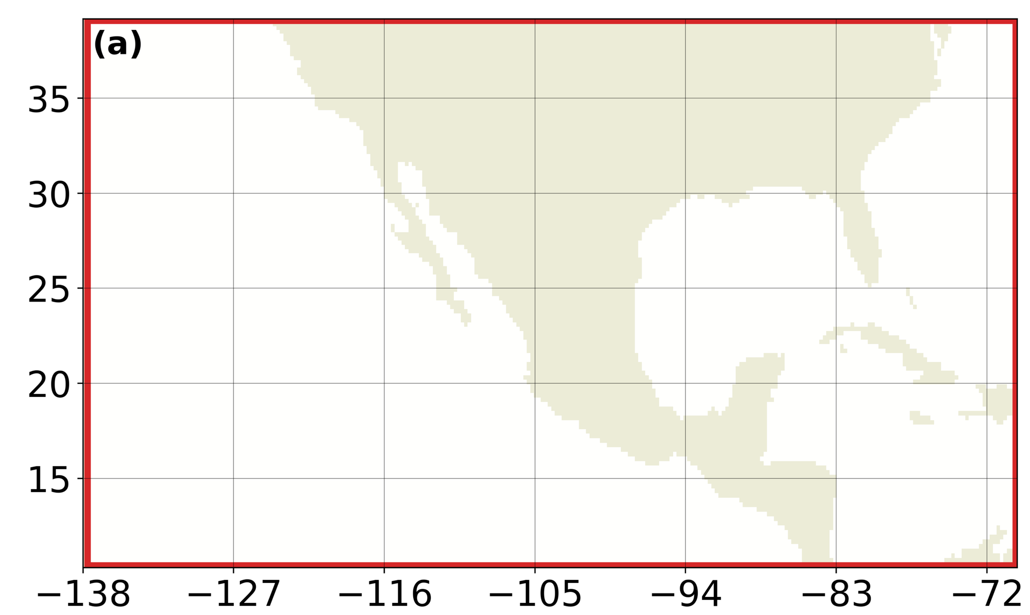
[2b] Compute surface and subsurface metrics

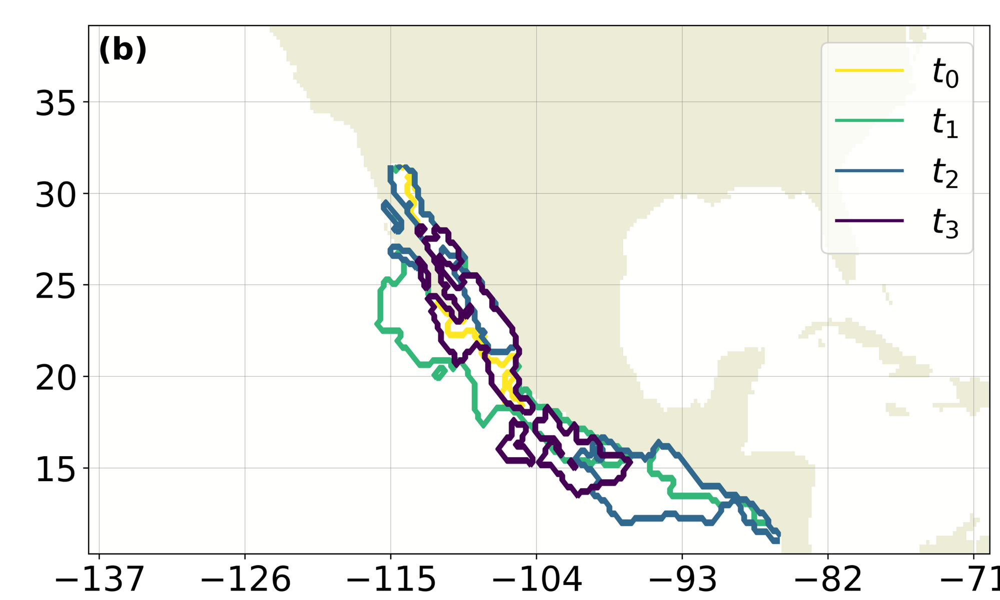


- Total events in MANSO: 911
- Key metrics computed:
  - Duration [days]
  - Area evolution [m²]
  - Volume evolution [m³]
  - Intensity evolution [°C]
  - Thickness [m]

[3] Diagnose closed WWMT budgets for 4D MHW masks using **xwmb**

a. Set up **xwmb**:
(in this study we set up two cases)





Case 1:
Computes a WWMT budget for the MANSO region

Case 2:
Computes a WWMT budget for individual spatiotemporally evolving features

b. Diagnose WMT budgets

$$\underbrace{\partial_t \mathcal{M}_{\geq}}_{\text{Mass change}} = \underbrace{\Psi_{\partial\mathcal{R}}}_{\substack{\text{horizontal}\\\text{mass transport}\\\text{into the region}}} + \underbrace{\mathcal{S}_{\Omega}}_{\substack{\text{external}\\\text{mass source}\\\text{(e.g. precipitation)}}} + \underbrace{\mathcal{G}_{\Omega}^{(\mathrm{T})}}_{\substack{\text{transformation}\\\text{processes related}\\\text{to heat uptake}}} - \underbrace{\mathcal{G}_{\Omega}^{(\mathrm{S})}}_{\substack{\text{Spurious}\\\text{Mixing}}}$$

- **WMT equation:** Quantify the processes (right-hand-side) contributing the the warm water mass tendency (left-hand-side)

🧰 **Tools:**

- MHW Tracker: `ocetrac` (github.com/ocetrac/ocetrac)
- WWMT evaluations: `xwmb` (github.com/hdrake/xwmb)